A Primer on the Inner Workings of Transformer-based Language Models

Javier Ferrando[1]*, Gabriele Sarti[2], Arianna Bisazza[2], Marta R. Costa-jussà[3]
[1]Universitat Politècnica de Catalunya, [2]CLCG, University of Groningen, [3]FAIR, Meta

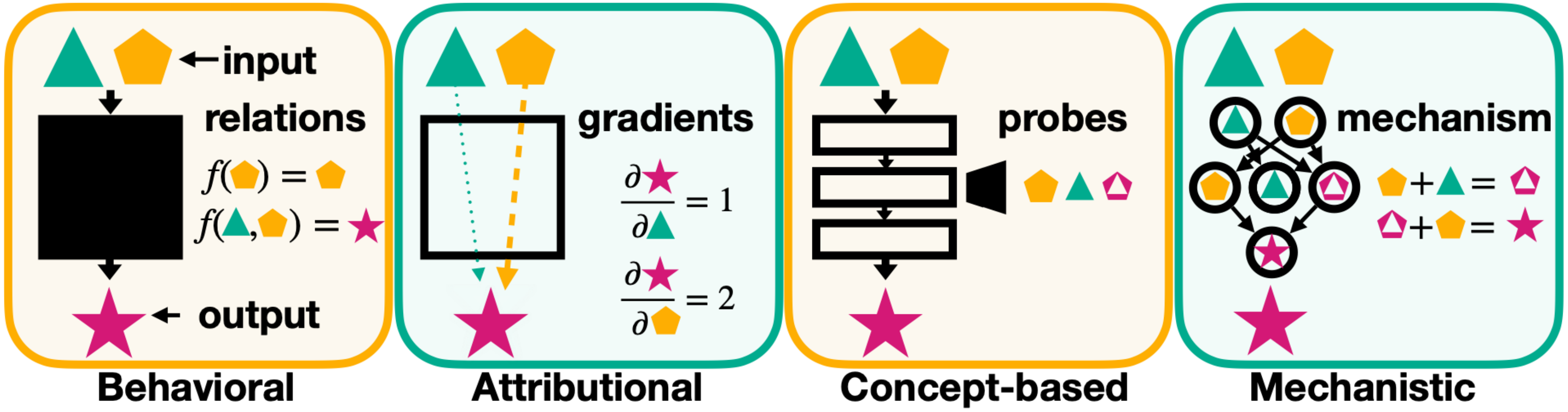
Part 1

HUMANE Lab, Jonghyeon Choi

preprint

2026.06.19

Background



Bereska, L., & Gavves, E. (2024). Mechanistic Interpretability for AI Safety — A Review. Transactions on Machine Learning Research.

Mechanistic Interpretability

Regular Computer Programs	Neural Networks
Reverse Engineering	Mechanistic Interpretability
Program Binary	Network Parameters
VM / Processor / Interpreter	Network Architecture
Program State / Memory	Layer Representation / Activations
Variable / Memory Location	Neuron / Feature Direction

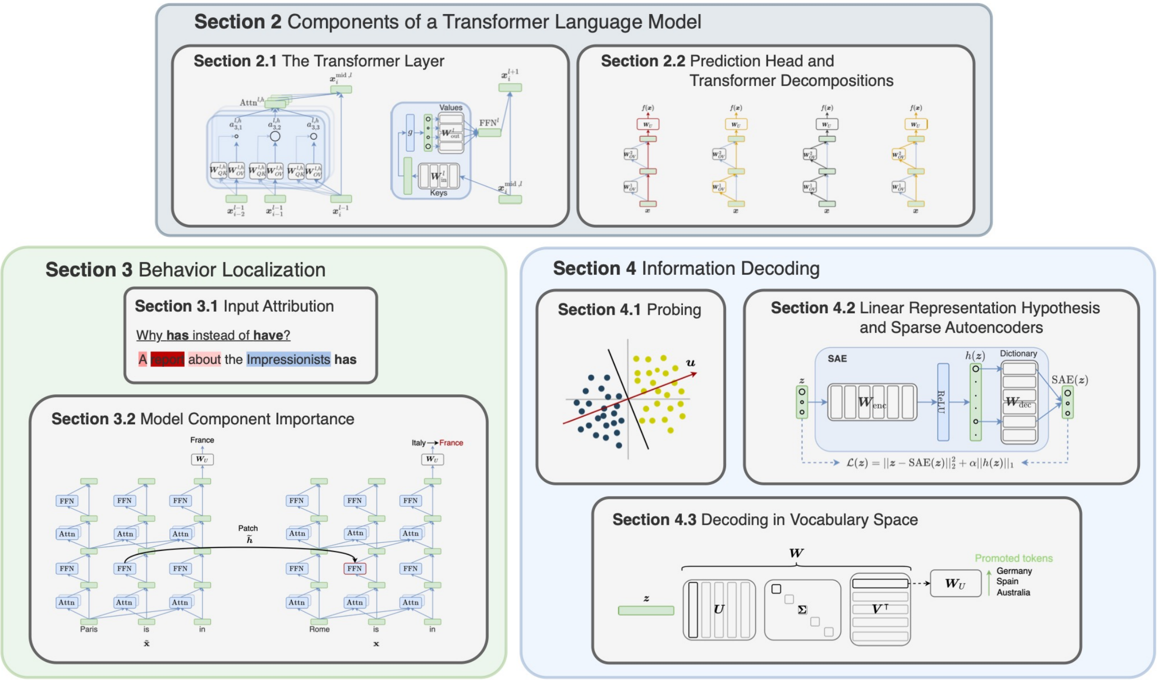
"Focuses on reverse-engineering neural networks into human-understandable algorithms"

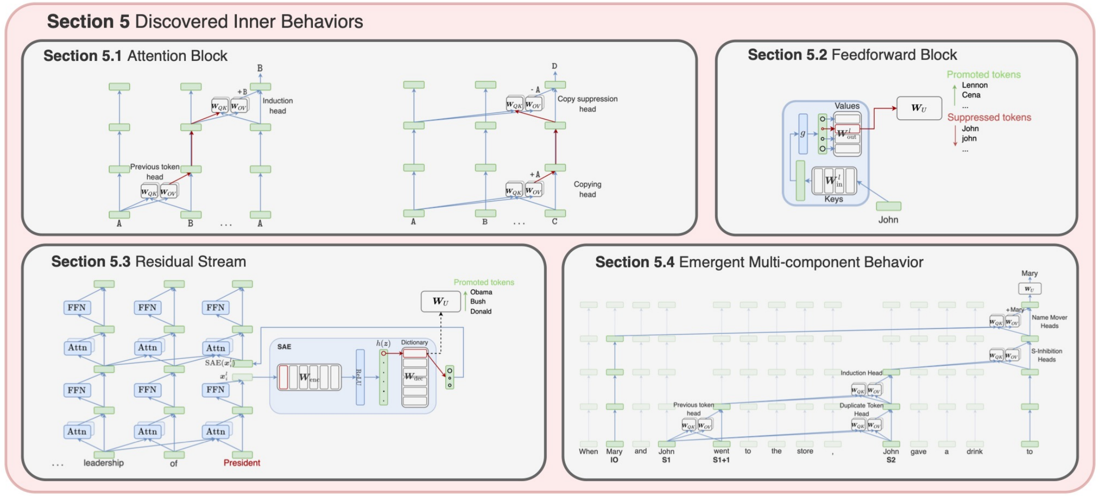
Olah, C. (2022). Mechanistic Interpretability, Variables, and the Importance of Interpretable Bases. Transformer Circuits.

Why Mechanistic Interpretability?

- Reliability
  - Correct output ≠ correct reasoning
  - Factual recall and hallucination
- Safety and fairness
  - Diagnose on biases and unsafe behavior
  - Finding refusal direction
- Model improvement
  - Understanding why it works → better models
  - e.g., interpretability-driven interventions

In this paper…

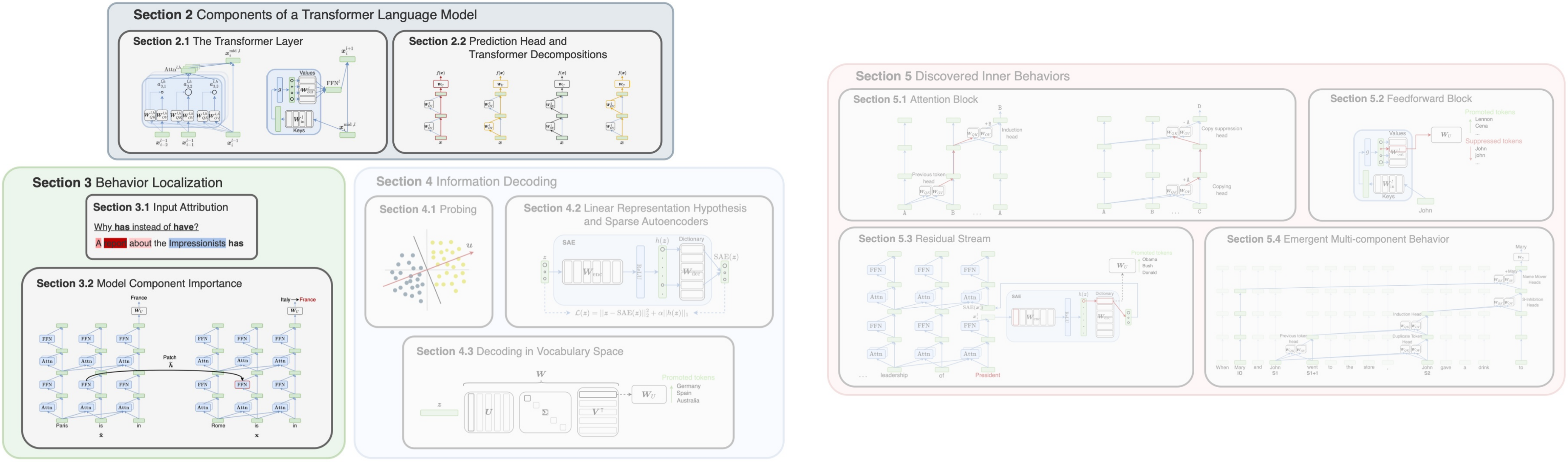




Methods

Insights

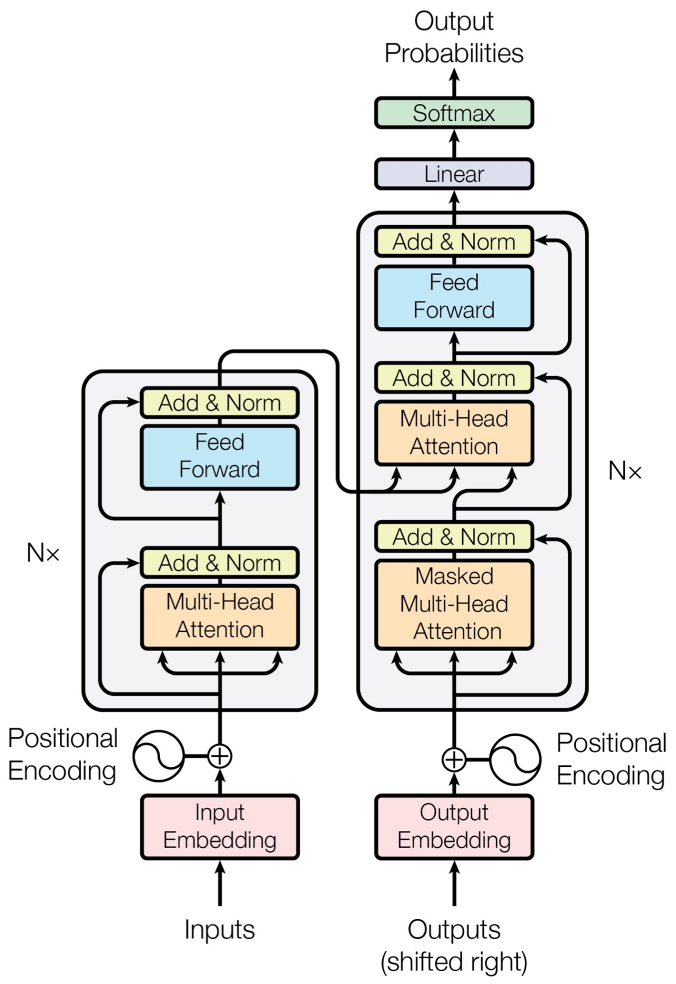
In this paper…



Methods

Insights

Understanding Transformer

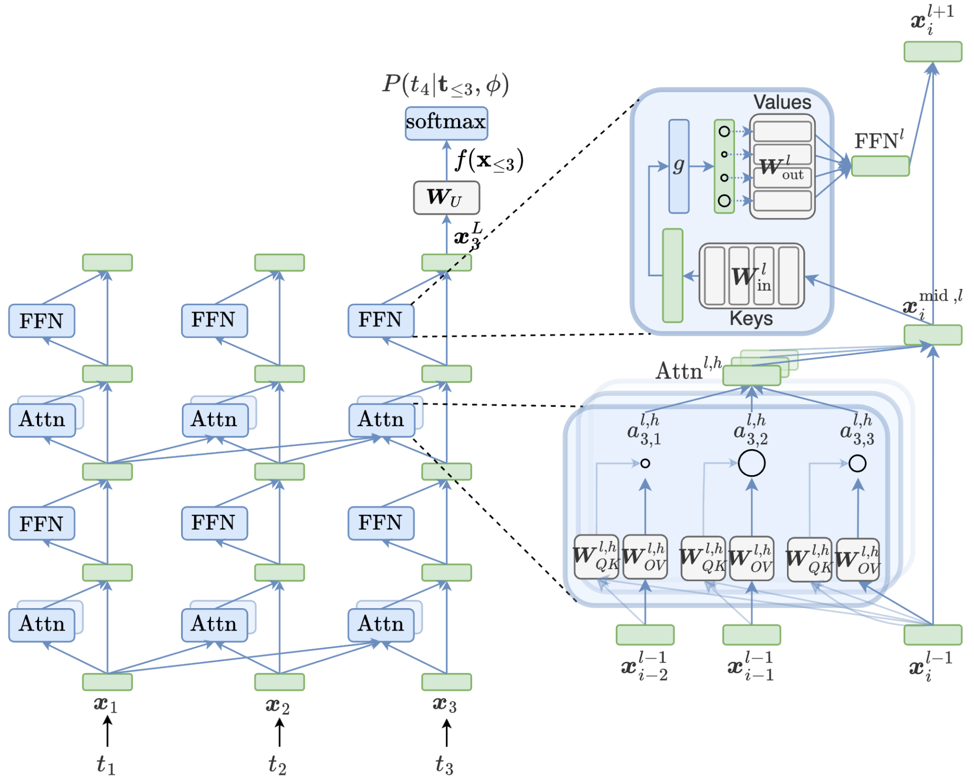


- Language model predicts next token based on p revious tokens
- Recent LLMs are decoder-only (auto-regressive)
- “Residual Stream”

$$P(t_1, \ldots, t_n) = P(t_1) \prod_{i=1}^{n-1} P(t_{i+1} | t_1, \ldots, t_i).$$

Vaswani, A. et al. (2017). Attention Is All You Need. NeurIPS.

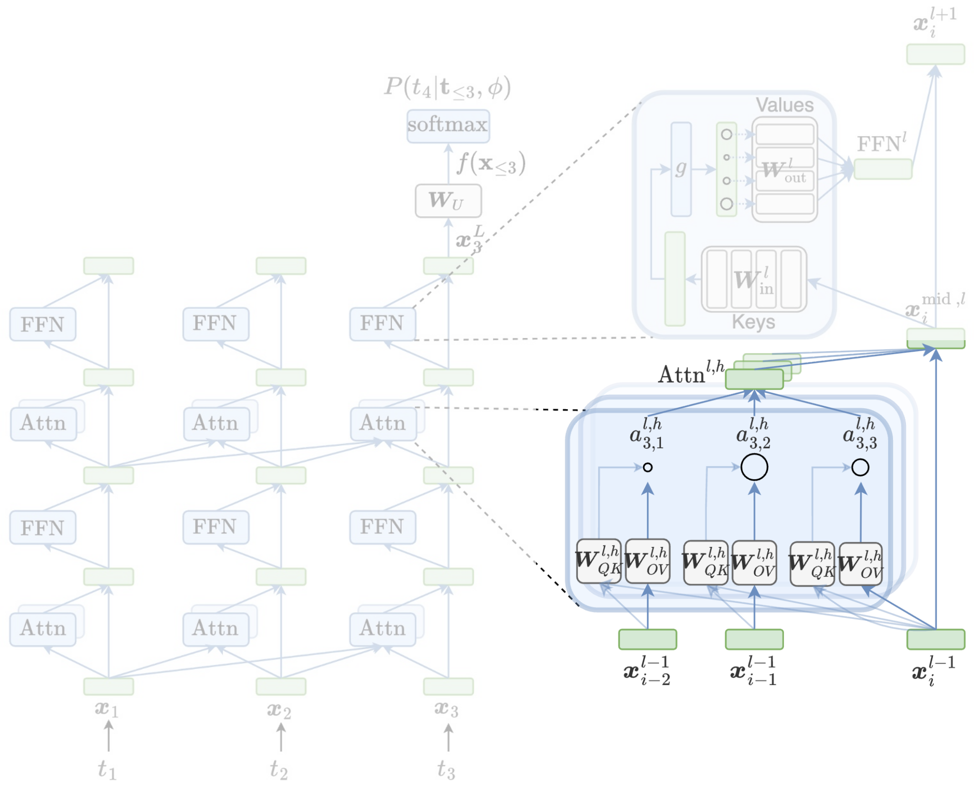
Understanding Transformer



Attention and FFN adds value in "residual stream"

Shared communication channel across components

Understanding Transformer (Attention)

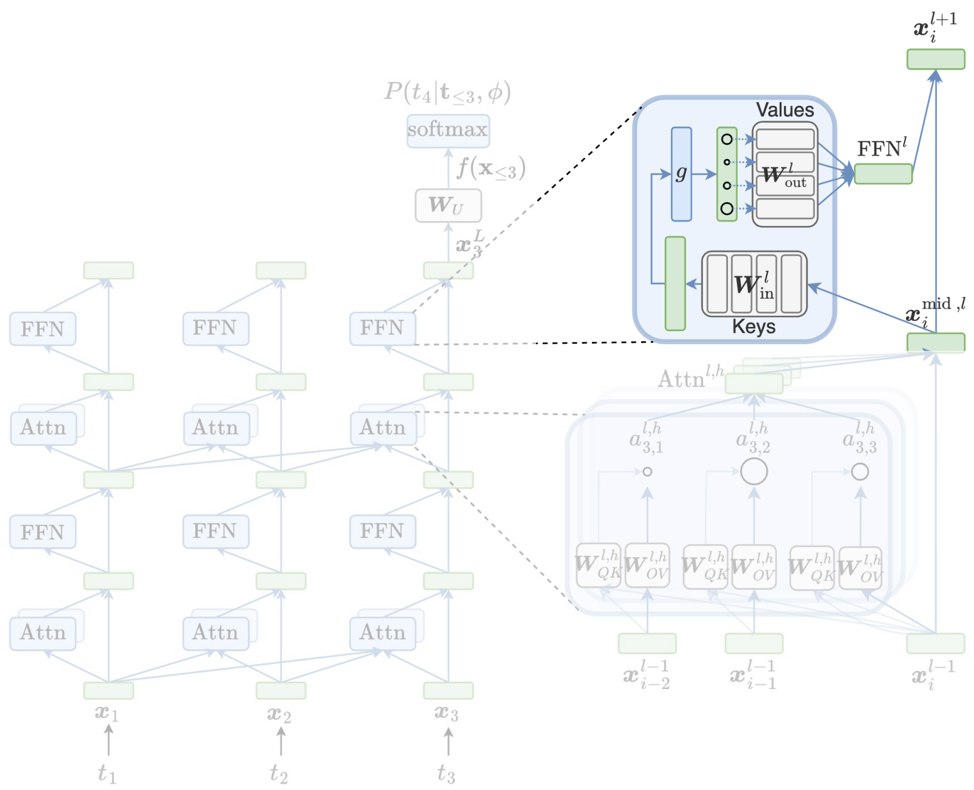


- QK circuit: computes attention weights — where to attend
- OV circuit: transforms attended token representations — writes to the residual stream

$$\mathrm{Attn}^l(\boldsymbol{X}_{\le i}^{l-1}) = \sum_{h=1}^{H} \mathrm{Attn}^{l,h}(\boldsymbol{X}_{\le i}^{l-1}),$$

$$\boldsymbol{x}_i^{\mathrm{mid},l} = \boldsymbol{x}_i^{l-1} + \mathrm{Attn}^l(\boldsymbol{X}_{\le i}^{l-1}).$$

Understanding Transformer (FFN)



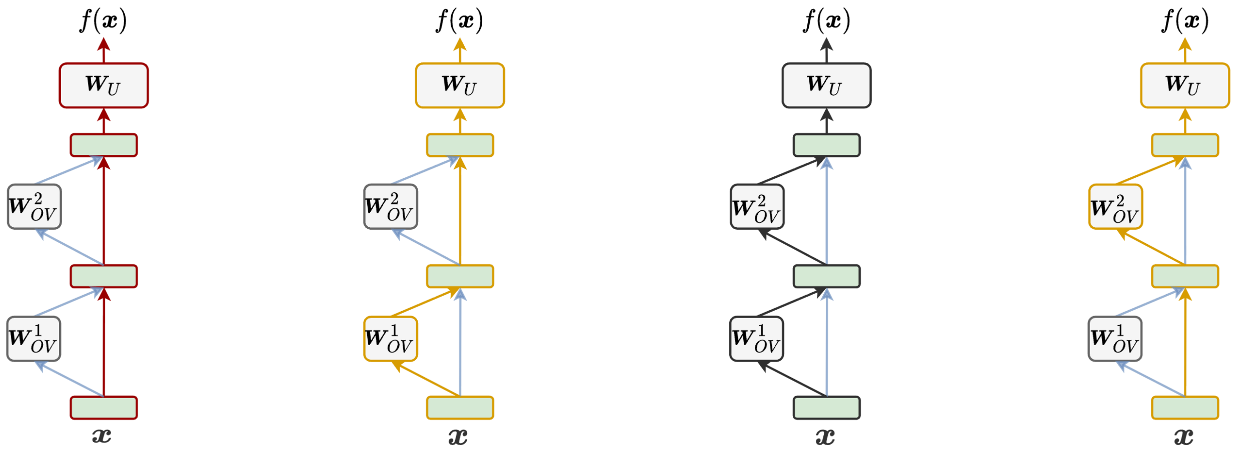
- Position-wise key-value memory
- W_{in} keys: detect features in the current residual stream
- W_{out} values: write activated feature updates back to the residual stream

$$\text{FFN}^l(\boldsymbol{x}_i^{\text{mid},l}) = \sum_{u=1}^{d_{\text{ffn}}} g_u(\boldsymbol{x}_i^{\text{mid},l}\boldsymbol{w}_{\text{in}_u}^l)\boldsymbol{w}_{\text{out}_u}^l$$

$$= \sum_{u=1}^{d_{\text{ffn}}} n_u^l \boldsymbol{w}_{\text{out}_u}^l,$$

Decomposition

- Last residual stream gets converted to next-token distribution of logits
- Every model component interacts through addition → linearity of residual stream
- Same as projecting each terms



$$
\begin{aligned}
f(\mathbf{x}) &= \boldsymbol{x}_n^L \boldsymbol{W}_U \\
&= \left(\sum_{l=1}^{L} \sum_{h=1}^{H} \mathrm{Attn}^{l,h}(\boldsymbol{X}_{\leq n}^{l-1}) + \sum_{l=1}^{L} \mathrm{FFN}^{l}(\boldsymbol{x}_n^{\mathrm{mid},l}) + \boldsymbol{x}_n \right) \boldsymbol{W}_U \\
&= \underbrace{\sum_{l=1}^{L} \sum_{h=1}^{H} \mathrm{Attn}^{l,h}(\boldsymbol{X}_{\leq n}^{l-1}) \boldsymbol{W}_U}_{\text{Attention head logits update}} + \underbrace{\sum_{l=1}^{L} \mathrm{FFN}^{l}(\boldsymbol{x}_n^{\mathrm{mid},l}) \boldsymbol{W}_U}_{\text{FFN logits update}} + \boldsymbol{x}_n \boldsymbol{W}_U .
\end{aligned}
$$

Behavior Localization

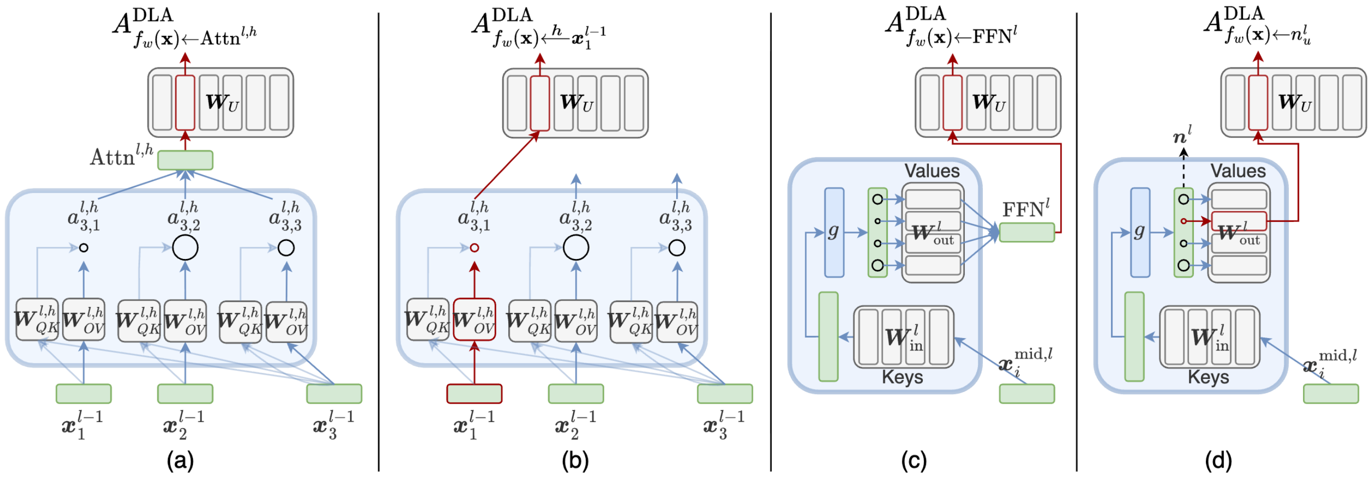
- What part of the model is responsible for this prediction?
- Input-level
  - What input tokens were important?
  - Input attribution
- Component-level
  - What components(e.g., head, FFN, neuron, etc.) were important?
  - Model Component Importance (e.g., DLA, patching, circuit analysis)

Behavior Localization: Input-level

- Gradient: sensitivity of the model to each element in the input
- Perturbation: changing input elements and see how it affects the results
- Context mixing: how information is mixed in attention

Behavior Localization: Component-level

- Direct Logit Attribution (DLA)
- How much a component increases or decreases the logit of a token



Attention | Attention Heads | FFN | Single Neuron

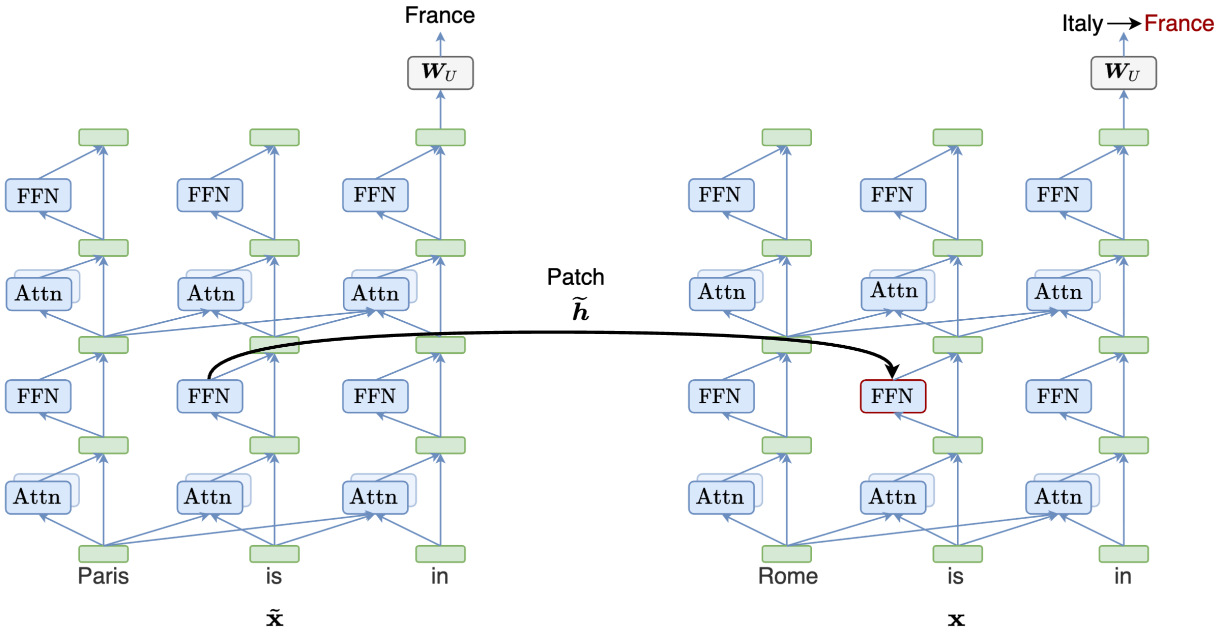
$$A^{\mathrm{DLA}}_{f_w(\mathbf{x})\leftarrow c} = f^c(\mathbf{x})\boldsymbol{W}_{U[:,w]},$$

$$A^{\mathrm{DLA}}_{f_w(\mathbf{x})\leftarrow n_u^l} = n_u^l \boldsymbol{w}^l_{\mathrm{out}_u}\boldsymbol{W}_{U[:,w]},$$

$$A^{\mathrm{DLA}}_{f_w(\mathbf{x})\xleftarrow{h}\boldsymbol{x}_j^{l-1}} = a^{l,h}_{n,j}\boldsymbol{x}_j^{l-1}\boldsymbol{W}^{l,h}_{OV}\boldsymbol{W}_{U[:,w]}.$$

Behavior Localization: Component-level

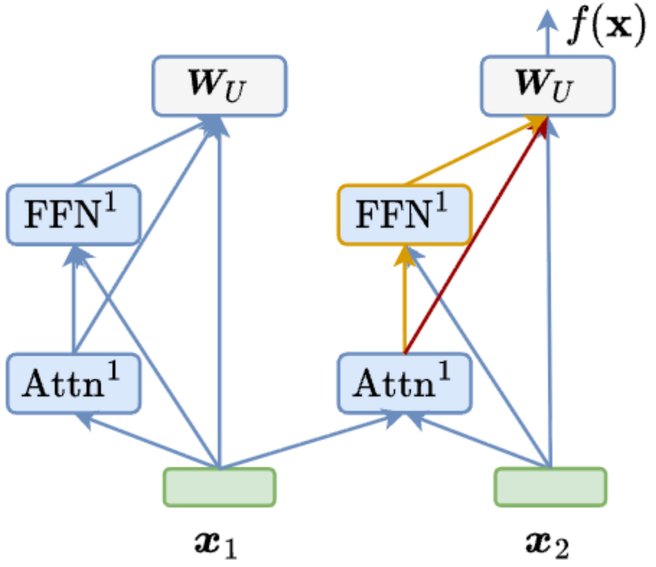
- Activation patching
- Changing activations to see if output changes (causal method)



$$A^{\text{Patch}}_{f(\mathbf{x})\leftarrow c} = \text{diff}(f(\mathbf{x}), f(\mathbf{x}|\text{do}(f^c(\mathbf{x}) = \tilde{\boldsymbol{h}}))).$$

Behavior Localization: Component-level

- Circuit analysis
- Subgraph of model components that work together to solve a task
- DLA and patching can identify important components



$$f(\mathbf{x}) = \overbrace{\mathrm{Attn}(\boldsymbol{X}_{\leq n})}^{\text{Attn direct path to logits}} \boldsymbol{W}_u + \mathrm{FFN}(\overbrace{\mathrm{Attn}(\boldsymbol{X}_{\leq n})}^{\text{Attn indirect path to logits via FFN}} + \boldsymbol{x}_n)\boldsymbol{W}_u + \boldsymbol{x}_n \boldsymbol{W}_u,$$

Behavior Localization

- Input-level
  - What input tokens were important?
  - Input attribution
- Component-level
  - What components(e.g., head, FFN, neuron, etc.) were important?
  - Model Component Importance (e.g., DLA, patching, circuit analysis)

Information Decoding

- What features or concepts are represented inside the model?
- To identify what information is encoded in hidden representations
- Residual stream, attention outputs, FFN activations, weights, etc.

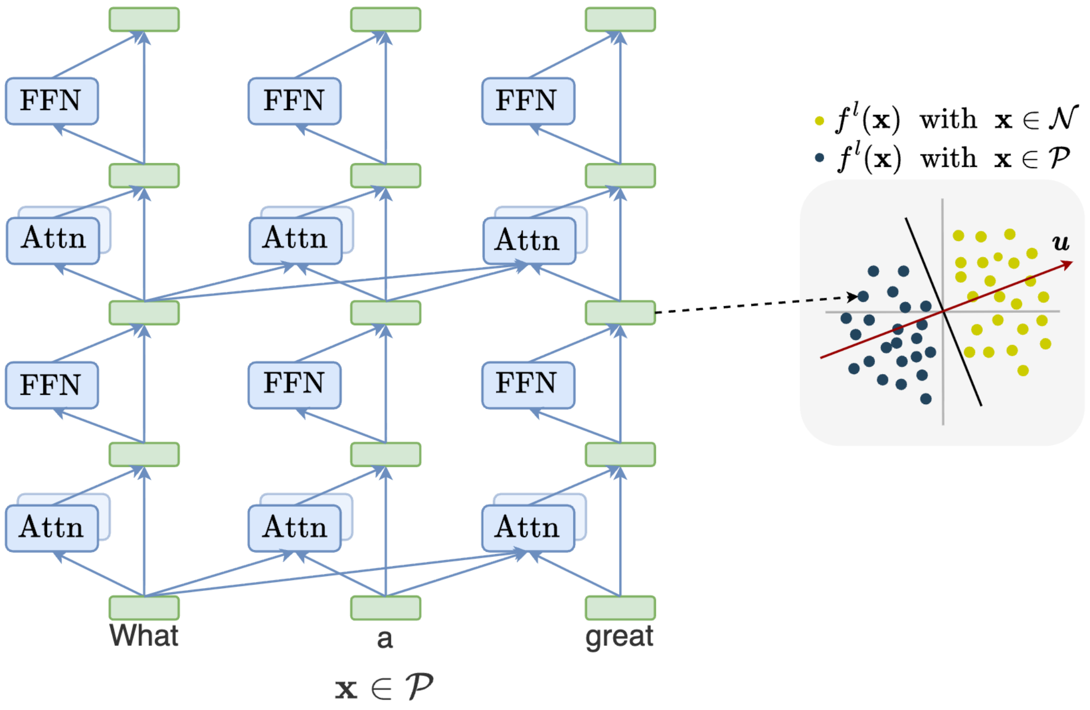
Information Decoding: Probing

- Probe: small supervised classifier trained on a model's internal representations to predict an input property
- Maps a hidden representation to a label

$$h = \mathrm{model}(x)$$

$$\hat{z} = \mathrm{probe}(h)$$

Information Decoding: Probing



- Probe trained to predict input sentiment with positive and negative

Information Decoding: Probing

- High probe accuracy → the property is decodable from the representation
- Powerful probe may learn the task itself rather than read existing informat ion
- Limitations of probing:
  - Probing is correlational, not causal
  - Encoded ≠ used by the model for its prediction

Conclusion of Part 1

- Goal and motivation of machine interpretation
- Residual stream as the foundation
- Behavior Localization (where) and Information Decoding (what)

Q&A